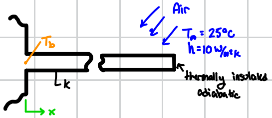
HW 5

1)



cylindrical pin fin $r = 0.001m$ $L = 0.05m$

$T_\infty = 25°C$ $h = 10 W/m^2K$ $k = 200 W/mK$

adiabatic tip: $-k \frac{dT}{dx}\Big|_{x=L} = 0$ $\frac{d\theta}{dx}\Big|_{x=L} = 0$ $\frac{\theta}{\theta_b} = \frac{\cosh(m(L-x))}{\cosh(mL)}$ $q_f = M\tanh(mL)$

$\theta = T - T_\infty$ $\theta_b = \theta(0) = T_b - T_\infty$ $m = \sqrt{\frac{hP}{kA_c}}$ $M = \theta_b\sqrt{hPkA_c}$

a) $T_b = 125°C$ find fin temp at $x = L/2$ and $x = L$

$\frac{\theta}{\theta_b} = \frac{\cosh(m(L-x))}{\cosh(mL)} = \frac{T - T_\infty}{T_b - T_\infty} \rightarrow T = \frac{(T_b - T_\infty)\cosh\left(\sqrt{\frac{hP}{kA_c}}(L-x)\right)}{\cosh\left(\sqrt{\frac{hP}{kA_c}}L\right)} + T_\infty$

$A_c = \pi(0.001m)^2 = 3.14\times10^{-6} m^2$ $P = 2\pi(0.001m) = 6.28\times10^{-3} m$

$m = \sqrt{\frac{(10 W/m^2K)(6.28\times10^{-3}m)}{(200 W/mK)(3.14\times10^{-6}m^2)}} = 10/m$

$T_b - T_\infty = 125 - 25 = 100K$ T_∞ in $K = 298.15K$

$T(L/2) = \frac{(100K)\cosh(10\cdot0.025)}{\cosh(10\cdot0.05)} + 298.15K \rightarrow \boxed{T(L/2) = 116.47°C}$

$T(L) = \frac{(100K)\cosh(0)}{\cosh(10\cdot0.05)} + 298.15K \rightarrow \boxed{T(L) = 115.68°C}$

b) for adiabatic tip: $\varepsilon_f = \sqrt{\frac{kP}{hA_c}}\cdot\tan(mL)$ $\eta_f = \frac{\tanh(mL)}{mL}$ from lecture notes

$\sqrt{\frac{kP}{hA_c}} = \sqrt{\frac{(200 W/mK)(6.28\times10^{-3}m)}{(10 W/m^2K)(3.14\times10^{-6}m^2)}} = 200$

$\varepsilon_f = (200)\tan((10/m)(0.05m)) \rightarrow \boxed{\varepsilon_f = 92.42}$

$\eta_f = \frac{\tanh(10\cdot0.05)}{10\cdot0.05} \rightarrow \boxed{\eta_f = 0.92}$ 0.92423

c) $\eta_f = \frac{\tanh(mL_c)}{mL_c}$ $L_c = L + D/4$

$L_c = 0.05m + \frac{0.001m}{2} = 0.0505$

$\eta_f = \frac{\tanh(10\cdot0.0505)}{10\cdot0.0505} \rightarrow \boxed{\eta_f = 0.92}$ 0.92285

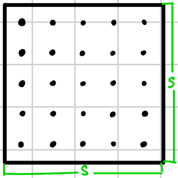
the results from (b) and (c) are very similar. While the result based on Table 3.5 is slightly smaller, they appear to be the same when rounded to 2 decimal decimal points, they are the same, making this a good approximation.

d) $R_f = \frac{1}{\eta_f h A_f}$

$A_f = LP = (0.05m)(6.28\times10^{-3}m) = 3.14\times10^{-4} m^2$

$R_f = \frac{1}{(0.92)(3.14\times10^{-4}m^2)(10 W/m^2K)} \rightarrow \boxed{R_f = 346.16\ K/W}$

1e)



$S = 3cm$

$q_{tot} = N q_f + q_{uf} = N \eta_f A_f \Theta_b + h A_{uf} \Theta_b$

$A_{uf} = (0.03m)^2 - A_c = (0.03m)^2 - 3.14 \times 10^{-6} m^2 = 8.97 \times 10^{-4} m^2$

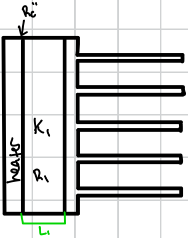
$\eta_o = 1 - \frac{NA_f}{A_{tot}}(1 - \eta_f) = 1 - \frac{(25)(3.14 \times 10^{-4} m^2)}{8.97 \times 10^{-4} m^2 + 3.14 \times 10^{-4} m^2}(1 - 0.92) \rightarrow \boxed{\eta_o = 0.48}$

$R_o = \frac{1}{\eta_o h A_{tot}} = \frac{1}{(0.48)(10 W/m^2K)(8.97 \times 10^{-4} m^2 + 3.14 \times 10^{-4} m^2)} \rightarrow \boxed{R_o = 172.03 K/W}$

$N q_f = N \eta_f A_f h \Theta_b = (25)(0.92)(3.14 \times 10^{-4} m^2)(10 W/m^2K)(100K) \rightarrow \boxed{N q_f = 72.22 W}$

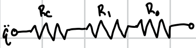
$q_{uf} = h A_{uf} \Theta_b = (10 W/m^2K)(8.97 \times 10^{-4} m^2)(100K) \rightarrow \boxed{q_{uf} = 8.47 W}$

f)



$L_1 = 0.02m$ $k_1 = 10 W/mK$ $R_c'' = 10^{-4} m^2K/W$

$q'' = 1000 W/m^2$



$R_c = R_c''/A = \frac{10^{-4} m^2K/W}{(0.03m)^2} = 0.11 K/W$

$R_1 = \frac{L_1}{k_1 A} = \frac{0.02m}{(0.03m)^2(10 W/mK)} = 2.22 K/W$

$q = q'' A = (0.03m)^2(1000 W/m^2) = 0.9W$ use T_B to find T_h

$\Delta T = R_q = (172.03 K/W + 0.11 K/W + 2.22 K/W)(0.9W) \rightarrow T_h - 398.15K = 2.097K \rightarrow \boxed{T_h = 127.097°C}$

$T_h - T_f = (172.03 + 0.11 + 2.22)(0.9) \rightarrow \Delta T = 127.097 K$

temperature is reduced significantly from previous design